DEI
DEPARTAMENTO
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MPI Basics
Collective Communication

Parallel and Distributed Computing

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Friday, 14 March 2025

NOS Award

This year, PDC is being sponsored by NOS.

At the end of the course:

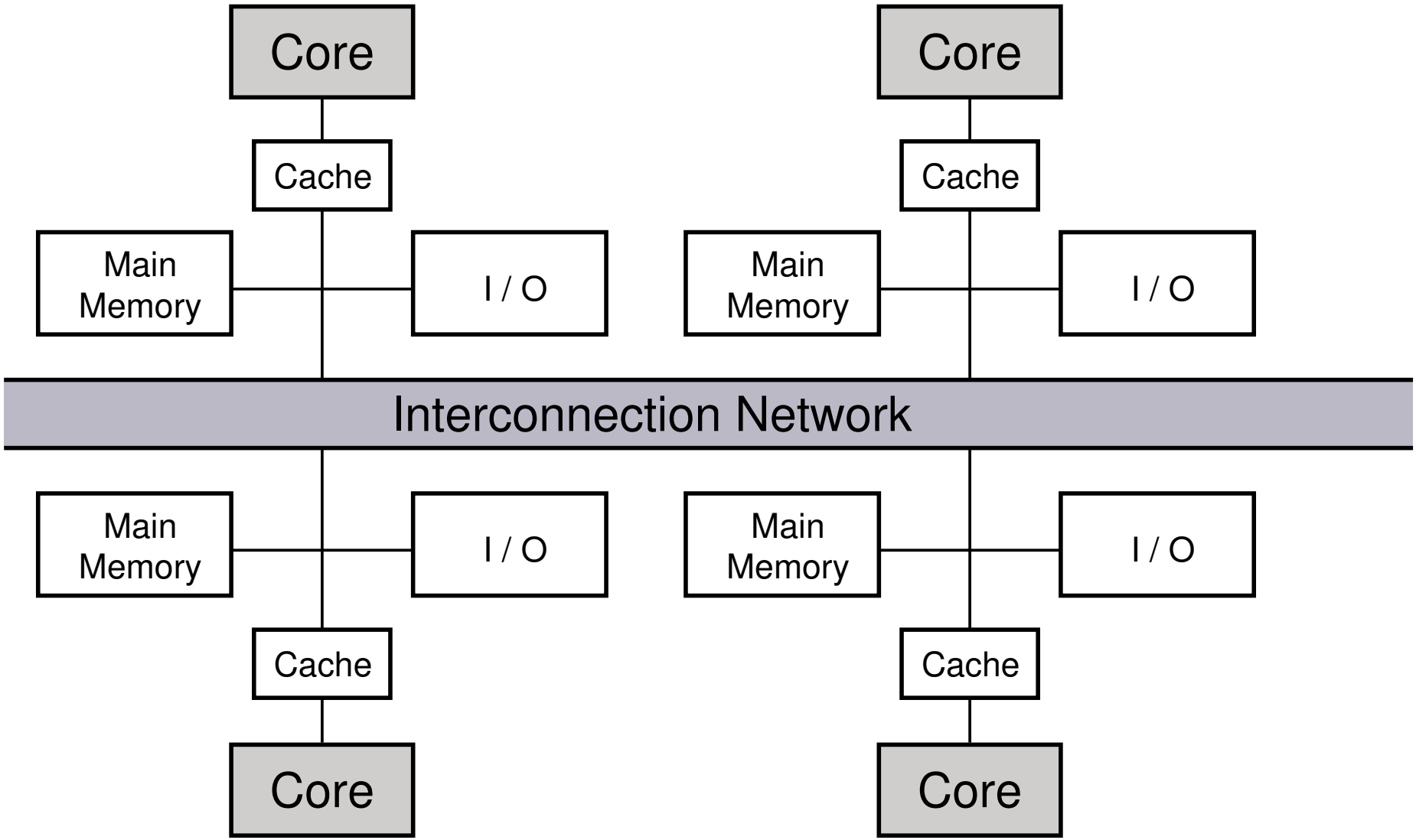
- 3 students with best grades will be selected to give a presentation about their project
- a jury will select the winner taking into account this presentation and the grade

Official information at IST's site.

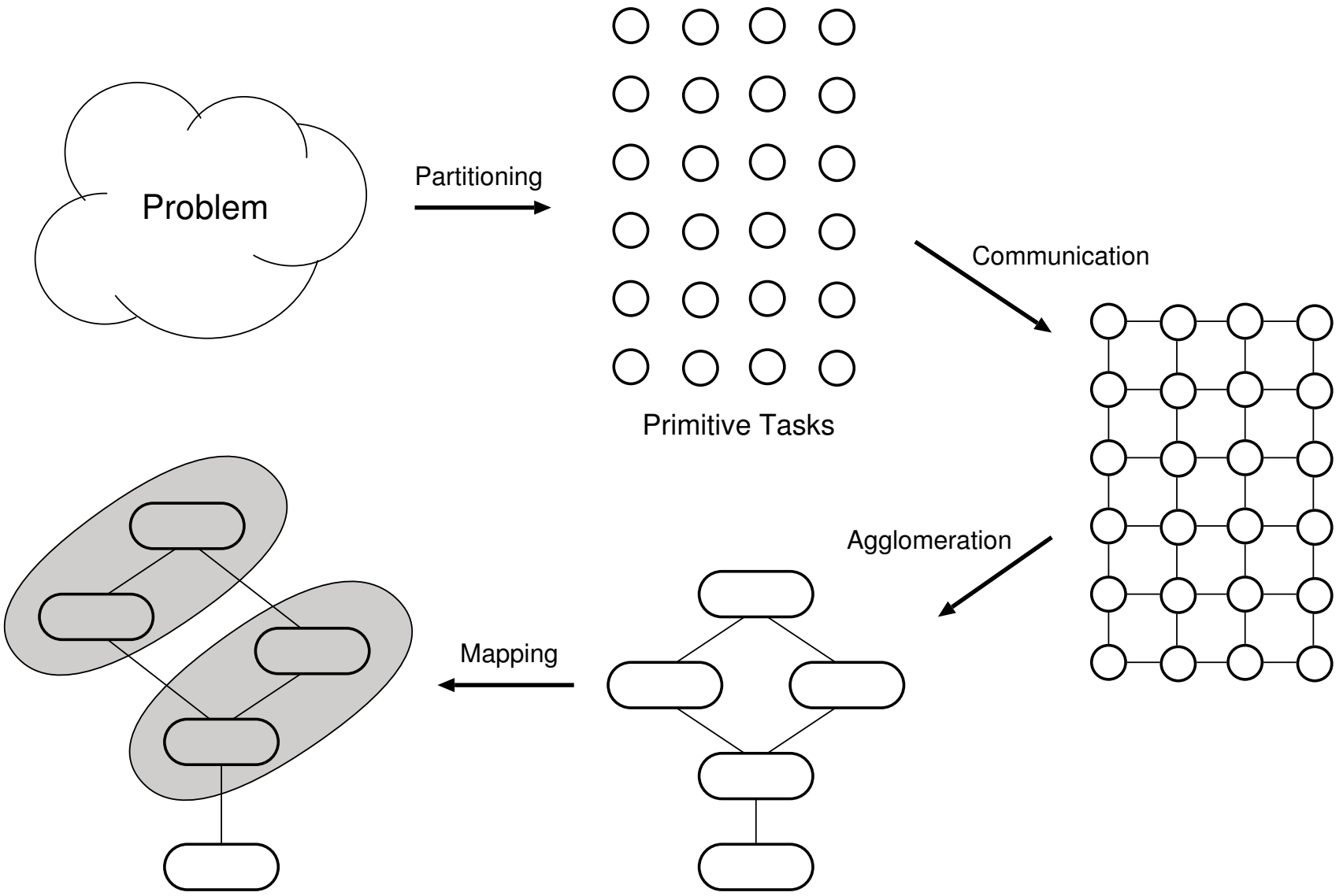
Outline

- MPI
  - Send/Receive: point to point messages
  - Asynchronous messages
  - Messages many-to-many
- Basic MPI application
  - Data decomposition options
  - Parallel algorithm development, analysis
  - Benchmarking
  - Optimizations

Distributed-Memory Systems



Foster's Design Methodology



Message Passing

- Message Passing: programming model for distributed memory parallel computers
  - Every processor executes an independent process
  - Disjoint address spaces, no shared data
  - All communication between processes is done cooperatively, through subroutine calls
- Message passing has succeeded because
  - Maps well to a wide range of hardware
  - Parallelism is explicit and communication is explicit
    - Forces the programmer to tackle parallelization from the beginning
  - MPI makes programs portable

What is MPI?

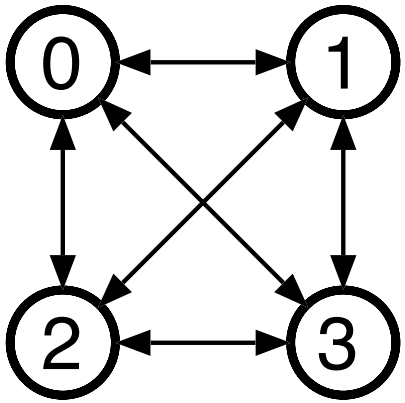
- Message Passing Interface (MPI) is the *de facto* standard for programming on distributed memory parallel computers
  - Library of routines that enable message passing applications
  - Interface specification, not a concrete implementation
- Before MPI
  - Different library for each type of computer
    - CMMD (Thinking Machines CM5)
    - NX (Intel iPSC/860, Paragon)
    - MPL (SP2) and many more …
  - PVM (Parallel Virtual Machine): tried to be a standard, but
    - Not high performance
    - Not carefully specified

MPI History

- MPI was developed by the MPI Forum, a voluntary organization representing industry, government labs and academia
  - MPI-1 (1994): codified existing practice
    - “Who am I?”, “how many processes are there?”
    - Send/recv communication
    - Collective communication e.g. broadcast, reduction, all-to-all
    - Lots of other stuff
  - MPI-2 (1997)
    - Parallel I/O
    - C++/Fortran 90
    - One-sided communication: get/put
  - MPI-3 (2012)
    - Dynamic process creation
    - Asynchronous collectives
    - Persistent collectives
    - Shared-memory windows
  - MPI-4 (2021)
    - Sessions model
    - Function versions (`_c`) with larger sizes
    - Process topologies
    - Improved error handler

An MPI Application

- The elements of the application are
  - 4 tasks, numbered `0` through `3`
  - Communication paths between them



- The set of tasks plus the communication channels is called "`MPI_COMM_WORLD`"

MPI “Hello, World”

```
#include <mpi.h>
#include <stdio.h>

main(int argc, char *argv[])
{
  int me, nprocs;

  MPI_Init(&argc, &argv);
  MPI_Comm_size(MPI_COMM_WORLD, &nprocs);
  MPI_Comm_rank(MPI_COMM_WORLD, &me);

  printf("Hi from node %d of %d\n", me, nprocs);

  MPI_Finalize();
}
```

Compiling and Running

- May be vary between machines, depends on actual MPI implementation
- OpenMPI is a high quality, open-source implementation of MPI
- Merger between well-known MPI implementations
  - FT-MPI, from the University of Tennessee
  - LA-MPI, from Los Alamos National Laboratory
  - LAM/MPI, from Indiana University
  - PACX-MPI, from University of Stuttgart
- Compile:

```
$ mpicc -o hello hello.c
```

- Start four processes:

```
$ mpirun -np 4 ./hello
```

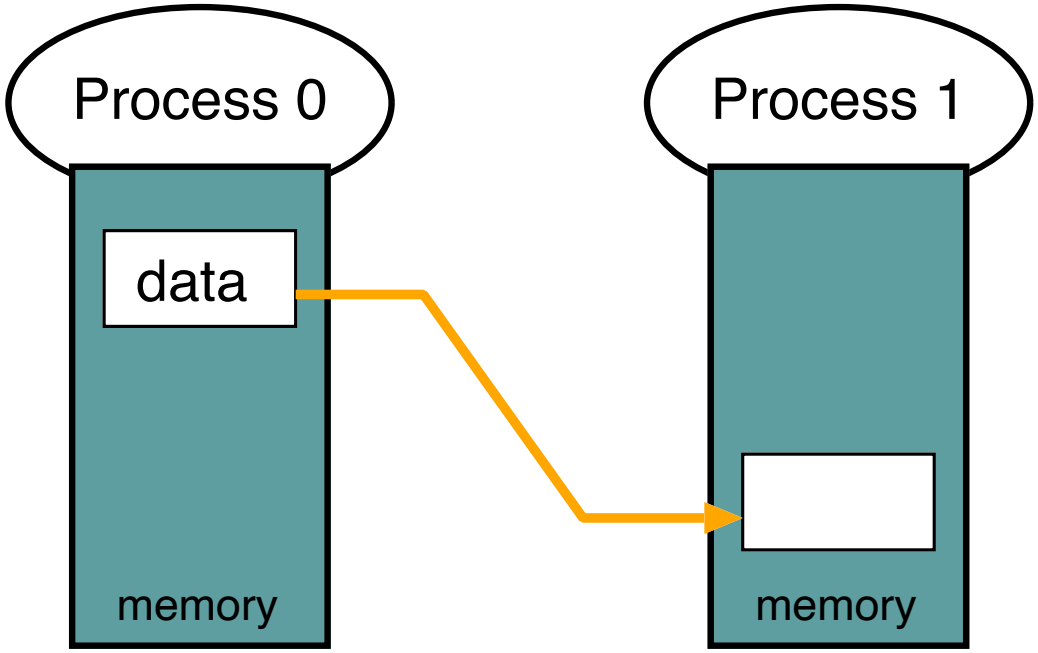
Sample Output

- Run with 4 processes

```
Hi from node 2 of 4
Hi from node 1 of 4
Hi from node 3 of 4
Hi from node 0 of 4
```

- Note that:
  - Order of output may change from run to run
  - Ability to use `stdout` is not guaranteed by MPI

Point-to-Point Communication




```
MPI_Send(data,...)
```

```
MPI_Recv(data,...)
```

Point-to-Point Example

- Task 0 sends array A to task 1, which receives it as B

Task 0

```
#define TAG 123
double A[10];
MPI_Send(A, 10, MPI_DOUBLE, 1, TAG, MPI_COMM_WORLD);
```

Task 1

```
#define TAG 123
double B[10];
MPI_Recv(B, 10, MPI_DOUBLE, 0, TAG, MPI_COMM_WORLD, &status);
```

or

```
MPI_Recv(B, 10, MPI_DOUBLE, MPI_ANY_SOURCE, MPI_ANY_TAG,
         MPI_COMM_WORLD, &status);
```

Source / Destination / Tag

- Send
  - Rank of task message is being sent to (destination)
  - Must be a valid rank (0,⋯ ,N –1) in communicator
- Receive
  - Rank of task message is being received from (source)
  - “Wildcard” `MPI_ANY_SOURCE` matches any source
- Tag
  - On the sending side, specifies a label for a message
  - On the receiving side, must match incoming message
  - On the receiving side, `MPI_ANY_TAG` matches any tag

Using Wildcards

- Unless there is a good reason to do so, do not use wildcards!
- However, there may be good reasons to use wildcards
  - Receiving messages from several sources into the same buffer (use `MPI_ANY_SOURCE`)
  - Receiving several messages from the same source into the same buffer, and don't care about the order (use `MPI_ANY_TAG`)

Pre-Defined Data Types

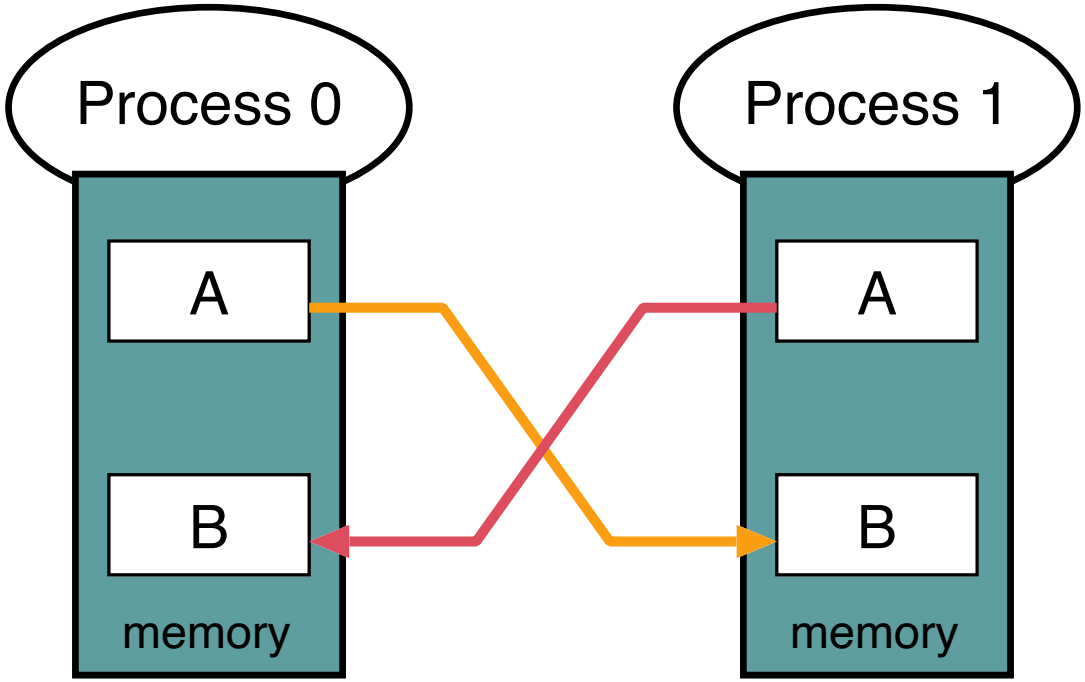
- C
  - `MPI_INT`
  - `MPI_FLOAT`
  - `MPI_DOUBLE`
  - `MPI_CHAR`
  - `MPI_LONG`
  - `MPI_UNSIGNED`
- Type-oblivious
  - `MPI_BYTE`
- MPI allows the definition of other data types/structures

Return Status

- `MPI_Status` is a structure
  - `status.MPI_TAG` is tag of incoming message (useful if `MPI_ANY_TAG` was specified)
  - `status.MPI_SOURCE` is rank of the sender of incoming message (useful if `MPI_ANY_SOURCE` was specified)
  - How many elements of given datatype were received

```
MPI_Get_count(MPI_Status *status,
              MPI_Datatype datatype,
              int *count)
```

Swapping Data




```
MPI_Send(A,...)
MPI_Recv(B,...)
```

```
MPI_Send(A,...)
MPI_Recv(B,...)
```

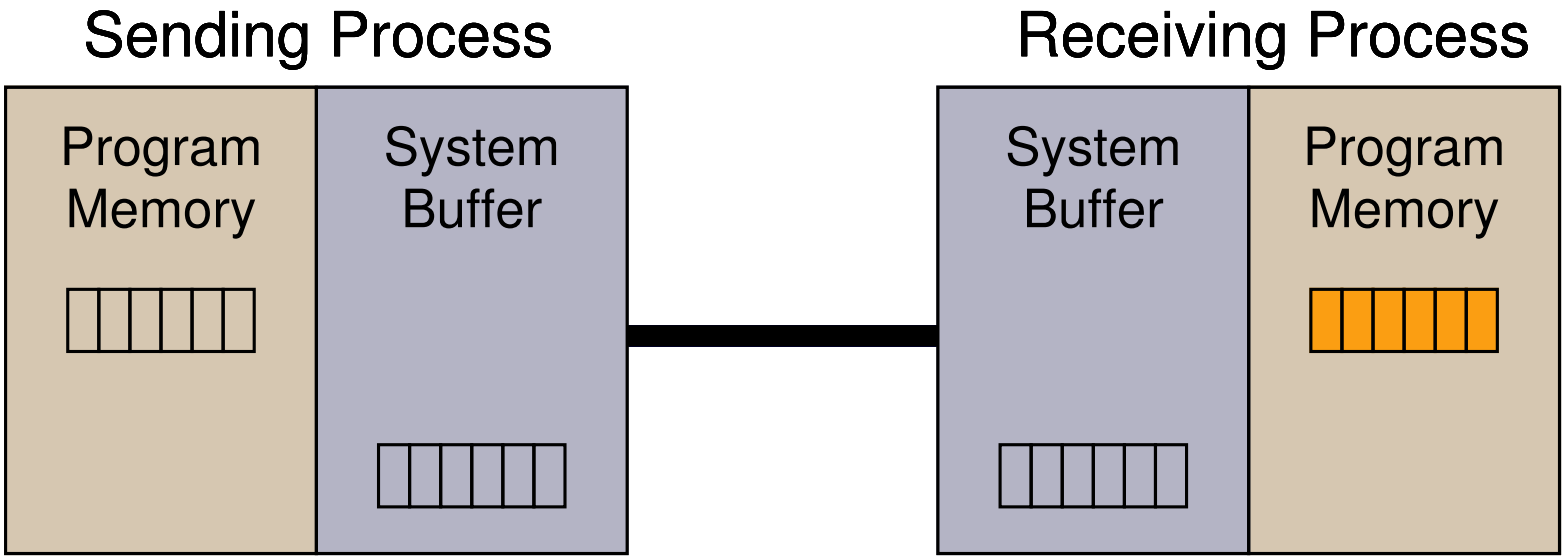
Blocking Operations

- `MPI_Send()` is blocking
  - Only returns when the send buffer can be reused
    - Data has been received by destination
    - Or MPI has saved the data somewhere

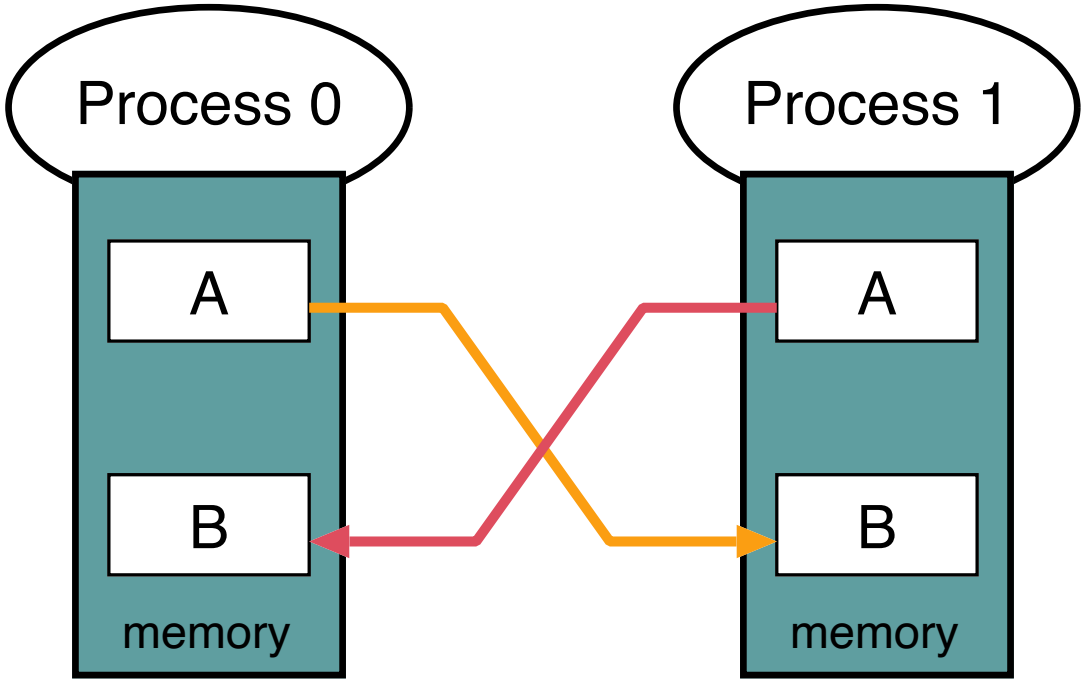
- `MPI_Recv()` is blocking
  - Only returns when the receive buffer has been filled with valid data

Does this work?

Internals of Send and Receive



Swapping Data




```
MPI_Send(A,...)
MPI_Recv(B,...)
```

```
MPI_Send(A,...)
MPI_Recv(B,...)
```

- Requires buffering to avoid deadlock!

  → a correct program does not rely on buffering!

Non-blocking Operations

```
#define MYTAG 123
#define WORLD MPI_COMM_WORLD
MPI_Request request;
MPI_Status status;
```

Task 0

```
MPI_Irecv(B, 100, MPI_DOUBLE, 1, MYTAG, WORLD, &request);
MPI_Send(A, 100, MPI_DOUBLE, 1, MYTAG, WORLD);
MPI_Wait(&request, &status);
```

Task 1

```
MPI_Irecv(B, 100, MPI_DOUBLE, 0, MYTAG, WORLD, &request);
MPI_Send(A, 100, MPI_DOUBLE, 0, MYTAG, WORLD);
MPI_Wait(&request, &status);
```

- No deadlock
- Data may be transferred concurrently

Overlapping Communication and Computation

- On some computers it may be possible to do useful work while data is being transferred: latency reduction

```
MPI_Request requests[2];
MPI_Status statuses[2];

MPI_Irecv(B, 100, MPI_DOUBLE, p, 0, WORLD, &request[1]);
MPI_Isend(A, 100, MPI_DOUBLE, p, 0, WORLD, &request[0]);

 .... do some useful work here ....

MPI_Waitall(2, requests, statuses);
```

  - `Irecv`/`Isend` initiate communication
  - Communication proceeds while processor is doing useful work
  - Hardware support necessary for true overlap

Operations on `MPI_Request`

- `MPI_Wait(INOUT request, OUT status)`
  - Waits for operation to complete
  - Returns information in `status`
  - Frees request object (and sets to `MPI_REQUEST_NULL`)
- `MPI_Test(INOUT request, OUT flag, OUT status)`
  - Tests to see if operation is complete
  - Returns information in `status` if complete
  - Frees request object if complete
- `MPI_Cancel(IN request)`
  - Cancels and completes a request
- `MPI_Waitall(..., INOUT array of requests, ...)`
- `MPI_Testall(..., INOUT array of requests, ...)`
- `MPI_Waitany/MPI_Testany/MPI_Waitsome/MPI_Testsome`

Non-blocking Communication Issues

- Obvious concerns
  - May not modify the buffer between **`Isend()`** and the corresponding **`Wait()`**
    - Results are undefined
  - May not look at or modify the buffer between **`Irecv()`** and the corresponding **`Wait()`**
    - Results are undefined
  - May not have two pending **`Irecv()`** for the same buffer
- Less obvious
  - May not *look* at the buffer between **`Isend()`** and the corresponding **`Wait()`**
  - May not have two pending **`Isend()`** for the same buffer

Information about Received Message

- It is possible to check information about an expected message before actually receiving it:

```
MPI_Probe(IN source, IN tag, OUT status)
```

  - Blocks until message is received
  - Returns information in `status`
  - Message not yet received, still requires `MPI_Recv`

- Non-blocking version:

```
MPI_Iprobe(IN source, IN tag, OUT flag, OUT status)
```

Communicators

- A communicator is an object that represents
  - A set of tasks
  - Private communication channels between those tasks

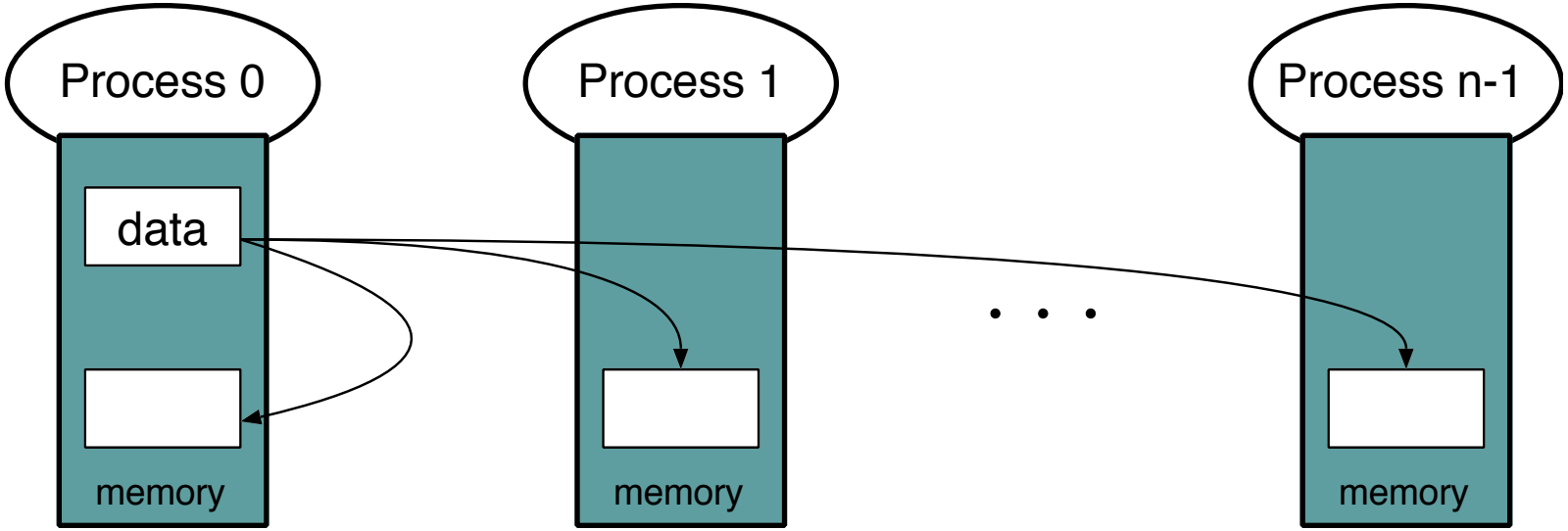
- `MPI_COMM_WORLD` is a communicator that includes all tasks and that is available at startup

- Communicators allow for the definition of the scope for collective operations

Collective Operations

- Collective communication is communication among a group of tasks
  - Broadcast
  - Scatter/gather
  - Global operations (reductions)
  - Parallel prefix (scan)
  - Synchronization (barrier)

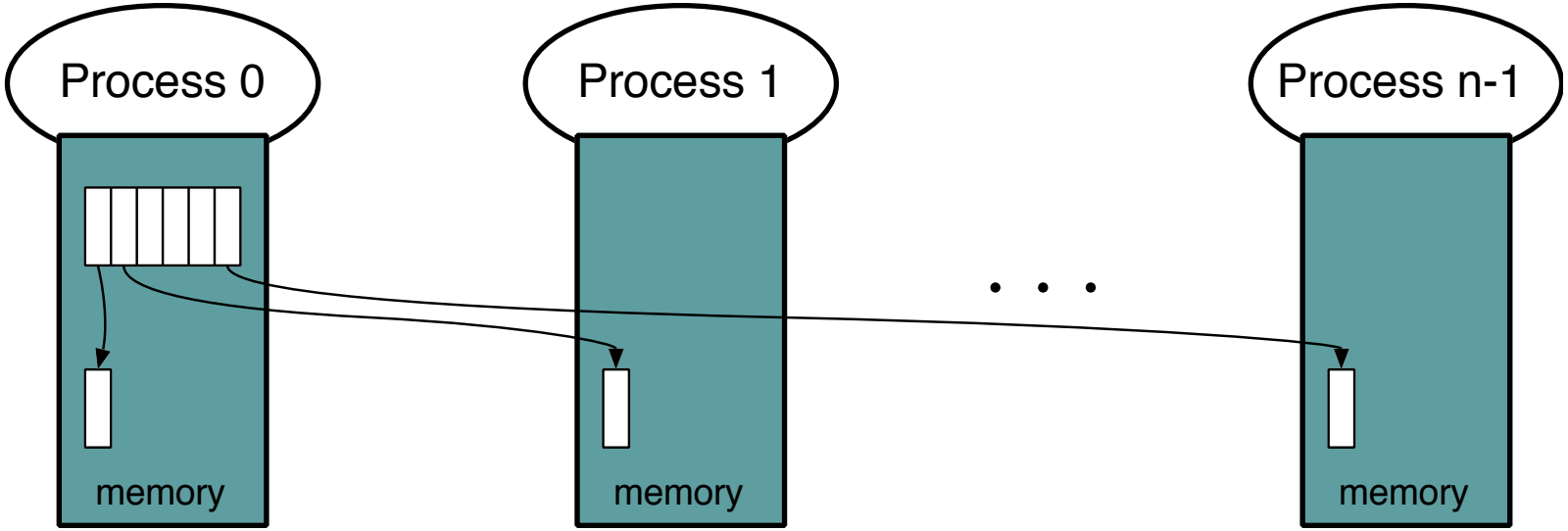
Broadcast




```
MPI_Bcast(buf, len, type, root, comm)
```

- Task with rank = `root` is source of data (in `buf`)
- Other tasks receive data

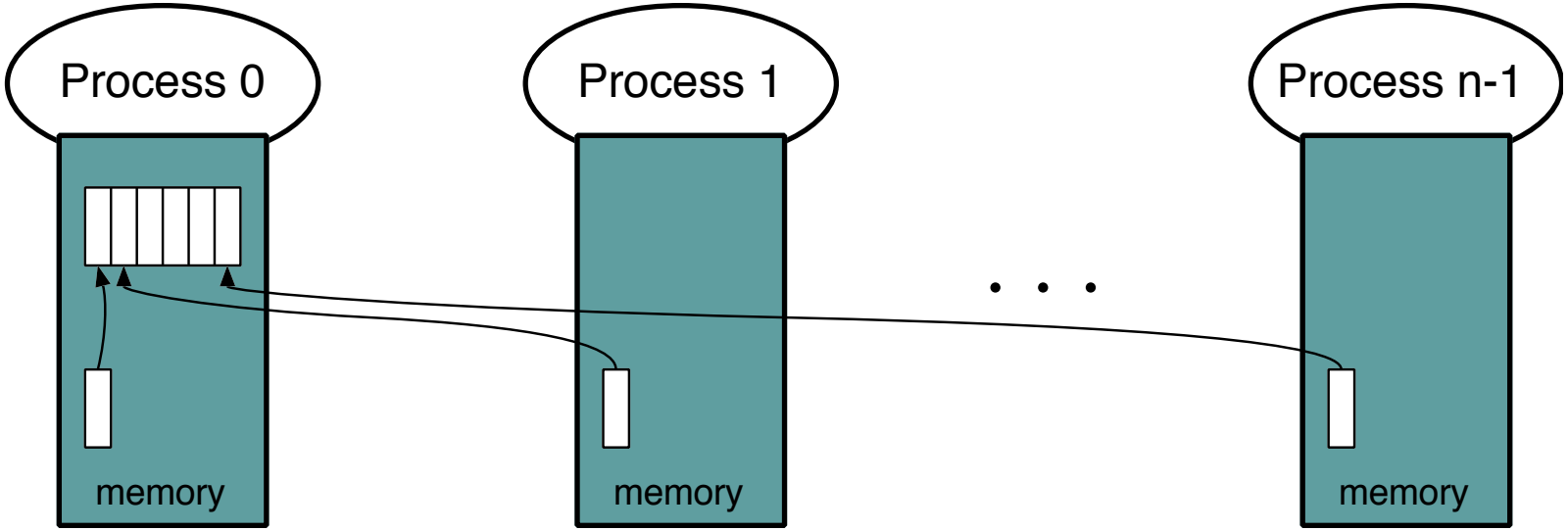
Scatter



`MPI_Scatter()` / `MPI_Scatterv()`

– Subparts of a single large array are distributed to tasks

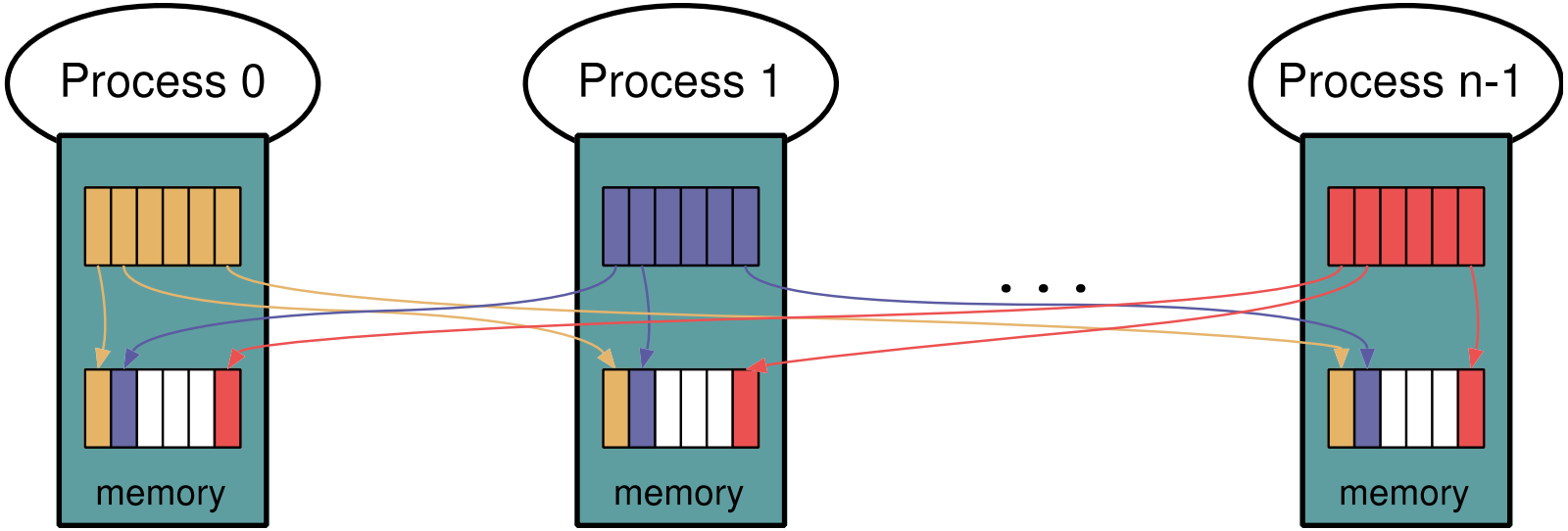
Gather




```
MPI_Gather() / MPI_Gatherv()
MPI_Allgather() / MPI_Allgatherv()
```

– Each task contributes local data that is gathered into a larger array

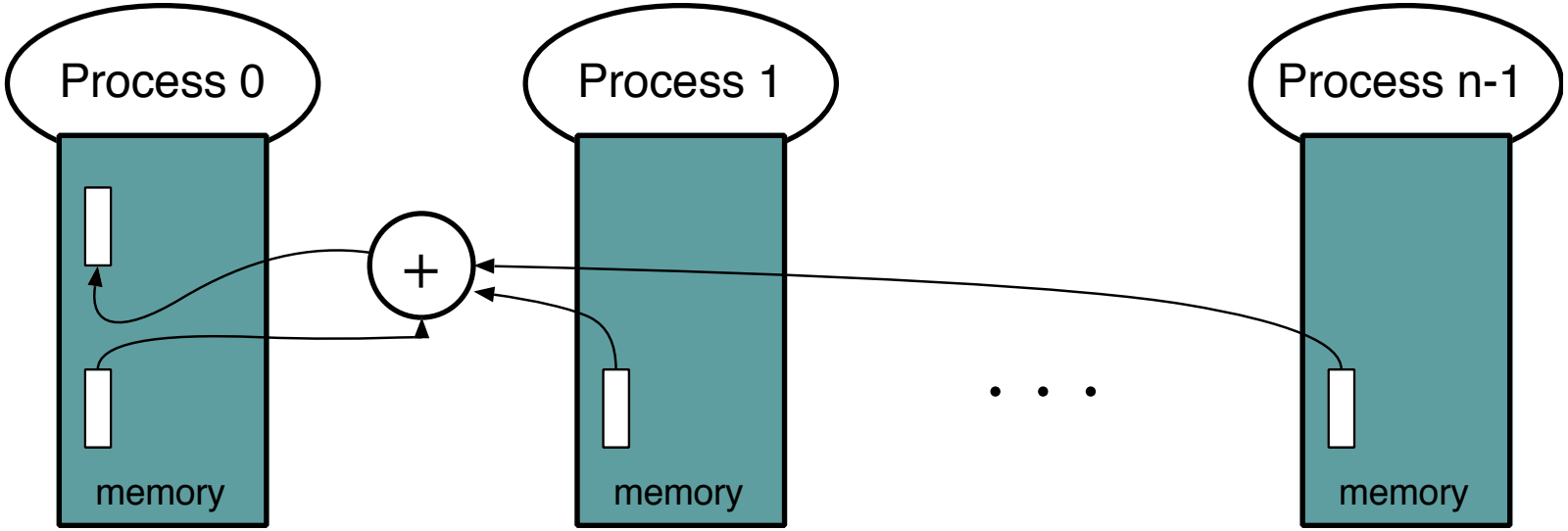
All to All Data Movements



`MPI_Alltoall(sndbf, sndct, sndt, rcvbf, rcvct, rcvt, comm)`

- All tasks send and receive data from all other tasks
- For a communicator with N tasks
  - `sndbf` contains P blocks of `sndct` elements each
  - `rcvbf` receives P blocks of `rcvct` elements each
  - each task sends block **`i`** of `sndbf` to task **`i`**
  - each task receives block **`j`** of `rcvbf` from task **`j`**

Reduction




```
MPI_Reduce(indata, outdata, count, type, op, root, comm)
MPI_Allreduce(indata, outdata, count, type, op, comm)
```

– Combine elements in input buffer from each task, placing result in output buffer.

Reduction

- Reduce: output appears only in buffer on root
- Allreduce: output appears on all processes

- Some operation types
  - `MPI_SUM`
  - `MPI_PROD`
  - `MPI_MAX`
  - `MPI_MIN`
  - `MPI_BAND`

- Arbitrary user-defined operations on arbitrary user-defined datatypes are possible

Reduction: Dot Product Example

```
/* distribute two vectors over all processes such that
   processor 0 has elements 0...99
   processor 1 has elements 100...199
   processor 2 has elements 200...299 etc. */

double dotprod(double a[100], double b[100])
{
  double gresult = lresult = 0.0;
  int i; /* compute local dot product */
  for(i = 0; i < 100; i++)
      lresult += a[i] * b[i];

  MPI_Allreduce(&lresult, &gresult, 1, MPI_DOUBLE, MPI_SUM,
                MPI_COMM_WORLD);

  return(gresult);
}
```

Synchronization

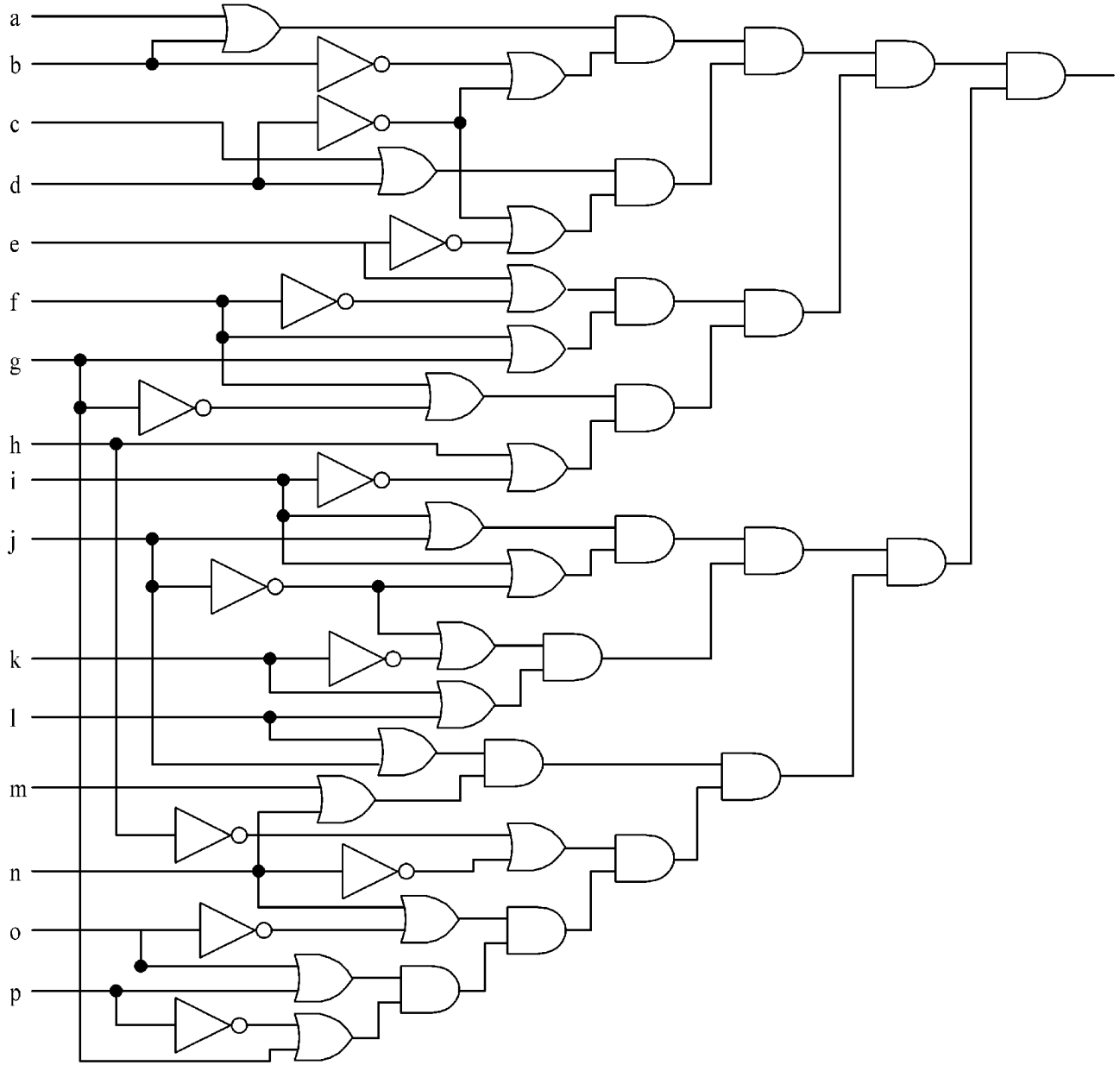
```
MPI_Barrier(communicator)
```

– No process leaves the barrier until all processes have entered it

First MPI Problem: Circuit Satisfiability

- Circuit Satisfiability (CSAT)
  - Determine a combination of input values that assert the output node of a logic circuit, or prove that no such combination is possible
  - Basic problem in VLSI circuit testing
  - NP-Complete problem!

CSAT Instance



CSAT Problem

- Determine all input combinations that assert the circuit output
  - Assert = make it assume logic value `1` (true)

- We will solve this problem through exhaustive search
  - Test all input combinations!

Foster's Methodology on CSAT

- Partitioning
  - Make each input combination test a primitive task
- Communication
  - No channels!
    - No interaction between tasks
- Aggregation and mapping
  - Distribute tasks evenly among available CPUs

C Program with MPI

```
#include <mpi.h>
#include <stdio.h>

int main (int argc, char *argv[])
{
  int i, id, p;

  MPI_Init (&argc, &argv);
  MPI_Comm_rank (MPI_COMM_WORLD, &id);
  MPI_Comm_size (MPI_COMM_WORLD, &p);

  for (i = id; i < 65536; i += p)
    check_circuit (id, i);

  printf ("Process %d is done\n", id);
  fflush (stdout);
  MPI_Finalize ();
  return 0;
}
```

C Program with MPI

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int main (int argc, char *argv[])
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  int i, id, p;

  MPI_Init (&argc, &argv);
  MPI_Comm_rank (MPI_COMM_WORLD, &id);
  MPI_Comm_size (MPI_COMM_WORLD, &p);

  for (i = id; i < 65536; i += p)
    check_circuit (id, i);

  printf ("Process %d is done\n", id);
  fflush (stdout);
  MPI_Finalize ();
  return 0;
}
```

C Program with MPI

```
#include <mpi.h>
#include <stdio.h>

int main (int argc, char *argv[])
{
  int i, id, p;

  MPI_Init (&argc, &argv);
  MPI_Comm_rank (MPI_COMM_WORLD, &id);
  MPI_Comm_size (MPI_COMM_WORLD, &p);

  for (i = id; i < 65536; i += p)
    check_circuit (id, i);

  printf ("Process %d is done\n", id);
  fflush (stdout);
  MPI_Finalize ();
  return 0;
}
```

C Program with MPI

```
/* Return 1 if 'i'th bit of 'n' is 1; 0 otherwise */
#define EXTRACT_BIT(n,i) ((n&(1<<i))?1:0)

void check_circuit(int id, int z) {
  int i, v[16]; /* Each element is a bit of z */
  for (i = 0; i < 16; i++)  v[i] = EXTRACT_BIT(z,i);
  if ((v[0] || v[1]) && (!v[1] || !v[3]) && (v[2] || v[3])
    && (!v[3] || !v[4]) && (v[4] || !v[5]) && (v[5] || !v[6])
    && (v[5] || v[6]) && (v[6] || !v[15]) && (v[7] || !v[8])
    && (!v[7] || !v[13]) && (v[8] || v[9]) && (v[8] || !v[9])
    && (!v[9] || !v[10]) && (v[9] || v[11]) && (v[10] || v[11])
    && (v[12] || v[13]) && (v[13] || !v[14]) && (v[14] || v[15])){
    printf("%d) %d%d%d%d%d%d%d%d%d%d%d%d%d%d%d%d\n", id,
      v[0],v[1],v[2],v[3],v[4],v[5],v[6],v[7],v[8],v[9],
      v[10],v[11],v[12],v[13],v[14],v[15]);
    fflush(stdout);
  }
}
```

Compiling and Running

- Compile command

```
$ mpicc <flags> -o <executable> <source>.c
```

  - Same flags as **gcc**, and more
  - Links with MPI libraries

- In our case

```
$ mpicc -O2 -o csat1 csat1.c
```

- Executing

```
$ mpirun -np <p> <executable> <args>
```

  - `<args>`: program's command-line arguments

Sample Output: 1 Task

```
$ mpirun -np 1 csat1
0) 1010111110011001
0) 0110111110011001
0) 1110111110011001
0) 1010111111011001
0) 0110111111011001
0) 1110111111011001
0) 1010111110111001
0) 0110111110111001
0) 1110111110111001
Process 0 is done
```

Sample Output: 2 Tasks

```
$ mpirun -np 2 csat1
0) 0110111110011001
0) 0110111111011001
0) 0110111110111001
1) 1010111110011001
1) 1110111110011001
1) 1010111111011001
1) 1110111111011001
1) 1010111110111001
1) 1110111110111001
Process 0 is done
Process 1 is done
```

Sample Output: 3 Tasks

```
$ mpirun -np 3 csat1
0) 0110111110011001
0) 1110111111011001
2) 1010111110011001
1) 1110111110011001
1) 1010111111011001
1) 0110111110111001
0) 1010111110111001
2) 0110111111011001
2) 1110111110111001
Process 1 is done
Process 2 is done
Process 0 is done
```

Deciphering the Output

- Output order only partially reflects order of output events inside parallel computer
  - If process A prints two messages, first message will appear before second
  - If process A calls **`printf`** before process B, there is no guarantee process A's message will appear before process B's message

Collective Communication

- Enhancing the program:

  write a new version of the Circuit Satisfiability program so that it returns the total number of solutions

- Modify function **`check_circuit`**
  - Return `1` if circuit satisfiable with input combination
  - Return `0` otherwise

- Each task keeps a local count of satisfiable input combinations it has found

- Incorporate sum-reduction into program

New Declarations and Code

```
int count;          /* Local sum */
int global_count; /* Global sum */
int check_circuit (int, int);

...

count = 0;
for (i = id; i < 65536; i += p)
  count += check_circuit (id, i)
```

Prototype of MPI Reduce

```
int MPI_Reduce (
  void          *operand, /* addr of 1st reduction element */
  void          *result,  /* addr of 1st reduction result */
  int           count,    /* reductions to perform */
  MPI_Datatype  type,     /* type of elements */
  MPI_Op        operator, /* reduction operator */
  int           root,     /* process getting result(s) */
  MPI_Comm      comm      /* communicator */
)
```

```
MPI_Reduce (&count, &global_count, 1, MPI_INT, MPI_SUM, 0,
            MPI_COMM_WORLD);
```

Only this task gets the result!

New Version of C Program

```
#include <mpi.h>
#include <stdio.h>

int main (int argc, char *argv[])
{
  int i, id, p;
  int count, global_count;

  MPI_Init (&argc, &argv);
  MPI_Comm_rank (MPI_COMM_WORLD, &id);
  MPI_Comm_size (MPI_COMM_WORLD, &p);

  count = 0;
  for (i = id; i < 65536; i += p)
    count += check_circuit (id, i);
  MPI_Reduce (&count, &global_count, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);
  printf ("Process %d is done\n", id);
  fflush (stdout);
  MPI_Finalize ();
  if (!id) printf("There are %d different solutions\n", global_count);
  return 0;
}
```

Sample Output of New Version: 3 Tasks

```
$ mpirun -np 3 csat2
0) 0110111110011001
0) 1110111111011001
1) 1110111110011001
1) 1010111111011001
2) 1010111110011001
2) 0110111111011001
2) 1110111110111001
1) 0110111110111001
0) 1010111110111001
Process 1 is done
Process 2 is done
Process 0 is done
There are 9 different solutions
```

Benchmarking Code

- Metric of interest: real time!

```
double MPI_Wtime()
```

Time in seconds since an arbitrary time in the past

```
double MPI_Wtick()
```

Timer resolution

- How to eliminate startup times?

```
int MPI_Barrier(MPI_Comm comm)
```

Barrier synchronization

Benchmarking Code

```
double elapsed_time;

...

MPI_Init (&argc, &argv);
MPI_Barrier (MPI_COMM_WORLD);
elapsed_time = - MPI_Wtime();

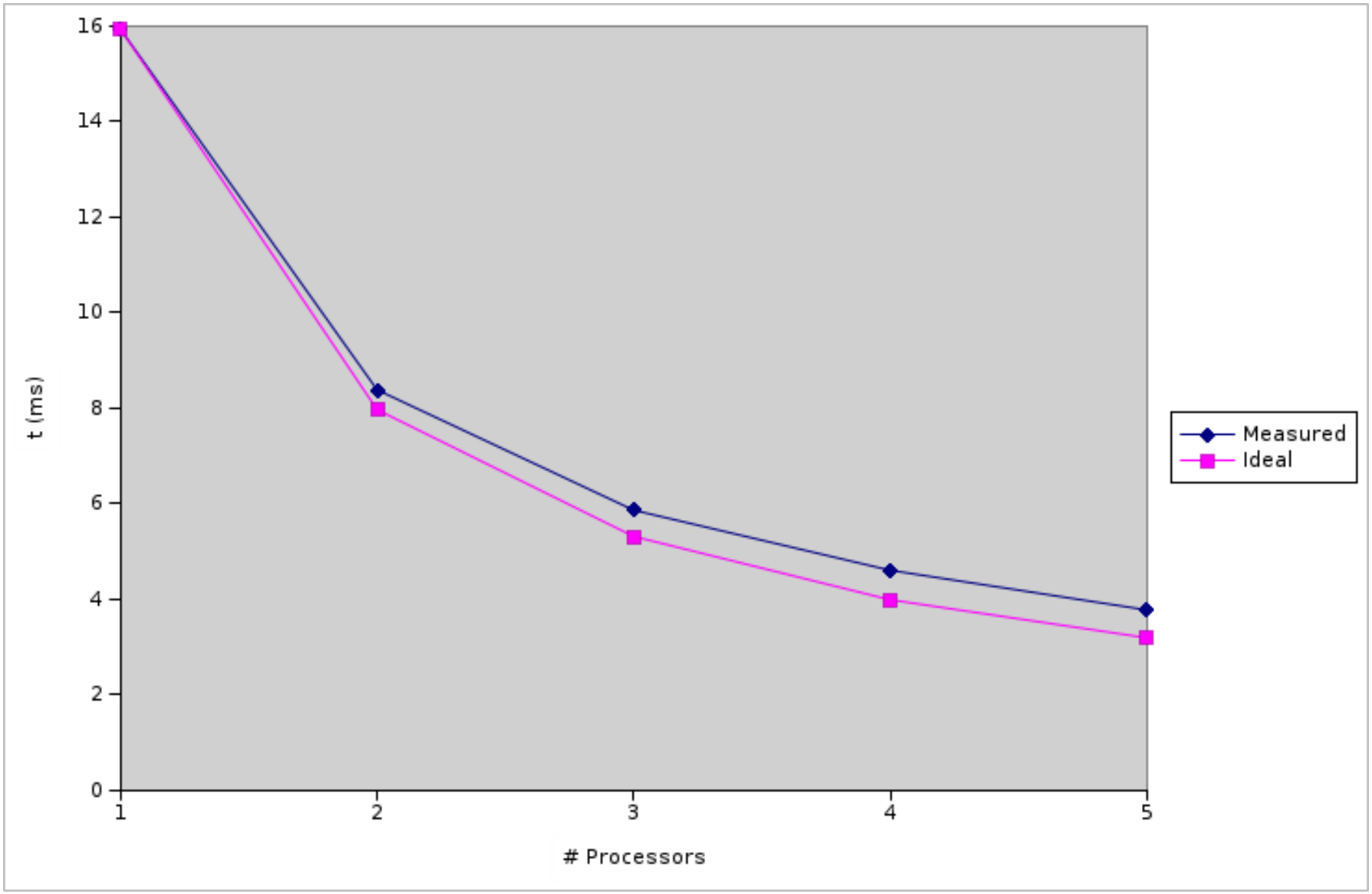
...

MPI_Reduce (...);
elapsed_time += MPI_Wtime();
```

Benchmarking Results

Processors	Time (ms)
1	15,93
2	8,38
3	5,86
4	4,60
5	3,77

Benchmarking Results



Review

- MPI
  - Send/Receive: point to point messages
  - Asynchronous Messages
  - Messages Many-to-many

- Basic MPI application
  - Data decomposition options
  - Parallel algorithm development, analysis
  - Benchmarking
  - Optimizations

Next Class …

- More complex application examples with MPI